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Tumpa Mahato

Current Affiliation

Indian Institute of Technology (IIT) Ropar — Punjab, India
Research Associate-I — Jan 2025- Present

- ANRF-CRG project "Twisted Links and Twisted Virtual Knot"
- Project Instructor: Dr. Madeti Prabhakar

Education

Indian Institute of Science Education And Research (IISER) Pune — Pune, India
Ph.D. in Mathematics — 2018-2024

- Advisor: Prof. Rama Mishra
- Thesis: Parameterization of Knotted Surfaces Arising from Classical and Welded Knots.

Sidho-Kanho-Birsha University — West Bengal, India
Masters in Mathematics — 2014-2016

- Percentage: 86.2

Research Interest

Knot Theory:
Surface knot theory, Knots on thickened surfaces: Virtual and Twisted knot theory, Welded knot theory, Low-dimensional topology, Hyperbolic knot theory

Work Experience

Shiv Nadar University — Greater Noida, India
Junior Research Fellow in a DST-SERB funded project — 2017-2018

- **Project title:** "Game-theoretic Approach of Modelling Antibiotic Drug Resistance as Self-Reinforcing Process by Social Factors and Economic Growth"

Publications

1. **Tumpa Mahato**, Ayaka Shimizu. Isolated Regions of a Link Projection. *Journal of Knot Theory and Its Ramifications*, Vol. 33, Issue No. 13, Article No. 2450042, 2024. https://doi.org/10.1142/S0218216524500421
2. **Tumpa Mahato**, Sahil Joshi, Rama Mishra, (In Press), Non-trivial Welded Knots and Ribbon Torus Knots, *Contemporary Mathematics*, Vol. on Knot Theory and Applications, American Mathematical Society. https://arxiv.org/abs/2404.00436v1
3. Aniruddha Deka, **Tumpa Mahato**, Samit Bhattacharyya. A mathematical model for distributed waning of vaccine-derived immunity: characterizing dynamical impact on measles elimination, *Franklin Open*, Vol. 9, 2024. https://www.sciencedirect.com/science/article/pii/S277318632400104X

Preprints

1. **Tumpa Mahato**, Prabhakar Madeti."Arc shift move and region arc shift move for twisted knots".(2026) https://arxiv.org/abs/2602.06432
2. Louis H. Kauffman, **Tumpa Mahato**, Rama Mishra. "Parametrizing Some Interesting Knotted Surfaces." (2025). https://doi.org/10.48550/arXiv.2507.19448
3. **Tumpa Mahato**, Rama Mishra, Polynomially Knotted 2-Spheres. https://arxiv.org/pdf/2307.07234

Manuscripts in preperation

1. **Tumpa Mahato**, Prabhakar Madeti, "Gordian complexes for twisted knots for arc shift and region arc shift move and the Q^z polynomial".

Research Highlights

1. **Wolfram Community Features**: The *Mathematica* notebook companion to "*Polynomially Knotted 2-spheres*" was officially selected and featured in the "Publication Materials" and "Staff Picks" editorial Columns by Wolfram. https://community.wolfram.com/groups/-/m/t/3122742

Invited Talks

1. **Parameterization of knotted surfaces arising from classical and welded knots**, International Conference on Differential Geometry and Topology, Jilin University, Changchun, China, Dec 12- Dec 15, 2024.
2. **Isolated-region number of a link projection**, Knots and Representation Theory, Online Seminar, Dec 2, 2024.
3. **Non-triviality and parameterization of knotted surface arising from one dimensioanl knots** , Knots and Representation Theory, Online Seminar, Oct 21, 2024.

Contributed Talks

1. **Arc shift move and region arc shift move for twisted knots**, Oberwolfach Seminar: Knots and their Interdisciplinary Applications, Mathematisches Forschungsinstitut Oberwolfach, Oberwolfach Germany, April 26, 2026 - May 1, 2026.
2. **Arc shift numbers and gordian complexes in twisted knot theory**, International Conference on "Knots and Related Topics", IIT Ropar, Punjab India, Dec 1-3, 2025.
3. **Parameterization of knotted spheres arising from 1-dimensional knots** , CYNOSURE 8.0 & National Symposium on Advances in Mathematics, IIT Ropar, Punjab, India, Feb 15 2025.
4. **Non-triviality and parameterization of ribbon torus knots**, Autumn School on Low- dimensional Topology and Related Topics, Institute for Basic Science - Center for Geometry and Physics, Pohang, South Korea, Sep 30- Oct 4, 2024.
5. **Parameterization and non-triviality of knotted surfaces arising from 1-dimensional knots,** Inter IISER NISER Math Meet, IISER TVM, Thiruvananthapuram, India, 2024.
6. **Parameterization of Knotted Surfaces**, Department Symposium, Department of Mathematics, Indian Institute of Science Education and Research Pune, India, 2023.

Workshop

1. **Oberwolfach Seminar: Knots and their Interdisciplinary Applications**, Mathematisches Forschungsinstitut Oberwolfach, Oberwolfach Germany, April 26, 2026 - May 1, 2026.
2. **Workshop on Low-dimensional Topology in collaboration with Max Planck Society and TIFR Mumbai**, IISER Pune, Pune, India, Dec 20 -Dec 29, 2023.
3. **Oberwolfach Seminar: Combinatorial and Geometric Knot Theory (hybrid meeting)**, Oberwolfach, Germany, Nov 21 -Nov 27, 2021.
4. **Knots through Web**, ICTS, Bangalore, India, Aug 24 - Aug 28, 2020.
5. **Volume Conjecture and Related Topics in Knot Theory**, IISER Pune, Pune, India, Dec 17-Dec 21, 2018.

Teaching

IIT Ropar, Punjab, India | Teaching Assistant — Jan 2025- Present
- Linear Algebra Course, Calculus for B.Tech students.

Savitribai Phule Pune University, Pune, India | Visiting Faculty — Sep 2024- Dec 2024
- Point Set Topology, Python, Sage for postgraduate M.Sc students

IISER Pune, Pune, India | Teaching Assistant — 2021
- Linear Algebra, Point set Topology

Awards and Honors

- **IAIAM Research fellowship** Indian Academy of Industrial and Applicable Mathematics 2024
- **Infosys travel grant** Infosys foundation 2024
- Cleared **National Eligibility Test (UGC)** with Junior Research Fellowship 2018
- Cleared **National Eligibility Test (CSIR)** with Junior Research Fellowship 2017
- **GATE Scholarship**: Awarded by National Coordination Board (NCB)-GATE, Department of Higher Education, Ministry of Human Resource Development (MHRD), Govt. of India 2017

Skills

Languages: English, Hindi, Bengali
Programming: `Python`, `MATLAB`, `SageMath`.
Softwares: `Mathematica`, LaTeX

References

1. **Prof. Rama Mishra**, Professor,
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2. **Dr. Madeti Prabhakar**, Associate Professor,
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3. **Dr. Tejas Kalelkar**, Associate Professor,
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Homepage: http://sites.iiserpune.ac.in/ tejas/

4. **Dr. Samit Bhattacharyya**, Associate Professor & Head, Department of Mathematics,
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